Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McKay, Christopher P Ph.D.		
Project Title:	Fluorometric Detection of Microorganisms on Sterilized Surfaces		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-6864
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/07/2001	**End Date:**	01/07/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Many important biomolecules fluoresce under Ultra-Violet (UV) light. Analysis of the fluorescence can indicate, number, type, and metabolic activity of microorganisms on surface. We propose to develop a non-contact sensor that using this principle to detect and characterize bacteria on surfaces with very low bacteria levels. The goal of this project is to develop a prototype flight instrument that can detect microorganisms using non-contact fluorometric techniques. Field testing of the unit in the extreme environments relevant to Astrobiology. In addition to detection analysis of the fluorescence signal can indicate number, type, and metabolic state of microorganisms. This work has applications to: 1) Astrobiology: field detection of microbes in extreme environments with low biomass; deserts and polar ice. 2) Flight instruments: Assay of microbial populations on Space Station surfaces, hardware, and biological experiments		

	3) Planetary protection: Determination of bioload on spacecraft surfaces on Earth (forward) and Mars (back contamination).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)